Kalhara Batangala

Lecturer (Probationary), Department of Electrical and Information Engineering
Faculty of Engineering, University of Ruhuna

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Research Interests

Wireless Communications • Optical and Fiber-Optic Networks • Free-Space Optical (FSO) and Satellite Communication • Computer Networks • Applied Machine Learning and Deep Learning

Profile

Computer Engineering graduate and probationary lecturer with a primary research interest in telecommunications, spanning wireless communications, optical/fiber networks, and free-space optical and satellite communication. Currently building a self-study foundation across these areas in preparation for postgraduate study (MSc) in telecommunications, with the longer-term aim of pursuing a PhD and an academic research career. Also brings applied experience in deep learning and computer vision, evidenced by a peer-reviewed conference publication and an ongoing manuscript in this area.

Education

B.Sc. Engineering (Hons.) in Computer Engineering, Second Class Upper Division Mar 2021 – Jul 2025
University of Sri Jayewardenepura *Sri Jayewardenepura, Sri Lanka*

- GPA: 3.31 / 4.00

G.C.E. Advanced Level & Ordinary Level 2005 – 2019
Kegalu Vidyalaya *Kegalle, Sri Lanka*

Academic Experience

Lecturer (Probationary) Aug 2026 – Present
Faculty of Engineering, University of Ruhuna *Galle, Sri Lanka*

- Deliver lectures in Computer Networking and Mobile Application Development for Electrical and Computer Engineering students.

Temporary Instructor Jan 2026 – Jul 2026
Faculty of Engineering, University of Ruhuna *Galle, Sri Lanka*

- Conducted and assisted laboratory sessions in the Department of Electrical and Information Engineering (DEIE), including Digital Logic Design and Digital Signal Processing labs.

Industry Experience

DevOps Engineering Intern Jun 2024 – Dec 2024
Sri Lanka Telecom PLC *Colombo, Sri Lanka*

- Collaborated with the frontend team on the SLT Portal and Internship Management System (React).
- Implemented CI/CD pipelines using Jenkins and Azure DevOps, reducing deployment time by 30%.

Publications

1. K. Batangala *et al.*, "NeuroSight: Brain Tumor Classification Using Deep Residual Networks," *accepted*, IEEE MERCon 2026.

Manuscripts in Preparation

1. K. Batangala *et al.*, "Cross-Task Feature Fusion for Multi-Task Womb Cancer Detection," manuscript in preparation.

Research Projects

NeuroSight – Brain Tumor Classification Using Deep Learning 2025

- VGG16/ResNet-based deep learning models for brain tumor classification from 2D MRI scans; work formed the basis of the MERCon 2026 publication.

Dog Breed Classification Model 2025

- Transfer-learning model (VGG16, TensorFlow) for image-based dog breed classification, using data augmentation to improve accuracy with limited training time.

Additional full-stack and mobile software projects (e-commerce, real-time sports scoring, AR-based interior design) are available on GitHub: github.com/KalharaBatangala.

Certifications & Professional Development

- Machine Learning, DeepLearning.AI (2024)
- Convolutional Neural Networks, DeepLearning.AI (2024)
- AWS Introduction to Cloud, AWS Educate (2025)
- Jenkins Level 1, KodeKloud (2025)
- Postman API Beginner, Postman Academy (2025)

University Service & Volunteering

- Vice President, CENSOC (2023/24)
- Organizing Committee, ENIGMA '24
- Organizing Committee, CodeSquad 4.0

Technical Skills

Programming & Tools: Python, PyTorch, TensorFlow, Scikit-learn, LaTeX, Linux
Domains: Computer Vision, Large Language Models, Retrieval-Augmented Generation (RAG), Agentic AI
Languages: Sinhala (native), English (fluent)

References

Dr. Thilina Weerasinghe
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University of Ruhuna
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Ms. Akarshani Amarasinghe
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